

Motivation | Problem Statement

Research Question

Scientific publication

- Long-distance BEV travel depends on en-route fast-charging, yet planning involves high uncertainty across decarbonization pathways (fleet size, driving range, modal shift)
- The challenge of ensuring sufficient geographic coverage while avoiding overcapacity from misaligned assumptions

RQ1

What are the required fast-charging capacities along Austria's highway network for battery-electric passenger cars in 2030 under different decarbonization scenarios, and where are these allocated?

Golab, A., Zwickl-Bernhard, S., & Auer, H. (2022). Minimum-cost fast-charging infrastructure planning for electric vehicles along the Austrian high-level road network. *Energies*, 15(6), 2147.
<https://doi.org/10.3390/en15062147>

- Broad BEV adoption requires a shift from home to public charging, while adoption needs to be considered beyond the early adopters that are characterized by high income
- Cross-regional spillover effects: local charging infrastructure expansion influences BEV adoption in neighboring regions

RQ2

- To what extent does the spatial density of public charging infrastructure influence BEV adoption across consumers with different income levels?
- How does a local rapid charging infrastructure expansion affect BEV adoption in neighboring regions with slower expansion?

Golab, A., Zwickl-Bernhard, S., & Auer, H. (2026). Public charger expansion: Impacts across income classes and beyond regional borders. *Transportation Research Part D: Transport and Environment*. In press.

passenger car fleets

commercial fleets

- Charging costs for battery-electric trucks vary geographically due to differences in electricity prices and network fees across countries
- Unknown magnitudes of potential charging infrastructure capacity allocations being affected by these differences

RQ3

What magnitude of charging infrastructure capacities is affected by geographically varying network fees and electricity prices in the cost-optimal planning of charging infrastructure allocation for long-haul battery-electric trucks?

Golab, A., Bakker, S., Zwickl-Bernhard, S., & Auer, H. (2026). Spatial distribution of charging patterns of inter-zonal long-haul battery-electric truck fleets. *Energy*. Submitted, March 2026.

- Estimating future charging demand from multiple commercial vehicle segments (light-duty trucks, heavy-duty trucks, buses) requires segment-specific modeling approaches due to their distinct operational patterns and geographic reach
- Fleet-projection-based methods miss transit freight traffic — vehicles passing through a country — which can account for a substantial share of charging demand

RQ4

What is the expected spatio-temporal distribution of charging demand from the commercial battery-electric vehicle fleet in Austria in 2040?

Golab, A., Loschan, C., Zwickl-Bernhard, S., & Auer, H. (2025). The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids. *Energy*, 316, 134385.
<https://doi.org/10.1016/j.energy.2025.134385>